of that was in 2007. However, Google then recruited the top engineers from the DARPA challenge to build a self-driving car which has met with some success.

The Google automobile is equipped with a plethora of sensors and computers to go about without crashing into people, but three key pieces of technology power this phenomenon. A combination of LIDAR (Light Detection And Ranging) and radar sensors continuously scan the environment to create a 3D map of the surroundings and calculate a distance to nearby objects. A video camera mounted on the dashboard looks forward to recognise other cars and pedestrians as well as to detect traffic signals. Lastly, sensors are installed in the rear wheels to measure the car's movements, thus gaining a perspective on its location and aiding in better manoeuvrability.

With all the technologies working together, Google's self-driving car has been tested to be quite road-safe so far. The only crash it has suffered recently has been a manual one.

Of course, Google is not the only one trying to build a car that drives itself. Germany's Free University of Berlin has a group that has been working exclusively on this aspect for over five years. The AutoNOMOS group recently travelled 80 kms in total as passengers during a test drive of their autonomous car "MadeInGermany". The car is driven by computers – the safety driver behind the steering wheel monitors only the car's behaviour. The autonomous car is a conventional VW Passat modified for "drive by wire". Electronic commands from the computer are handed over directly to the accelerator, the brakes and the steering wheel of the vehicle. Multiple sensors integrated in the car's chassis provide information about all cars or persons on the street. They allow the car to avoid obstacles, adjust its speed, or change lanes whenever necessary. The vehicle was tried out in full Berlin traffic and covered a 20km distance four times without any incidents.

MadeInGermany is the first autonomous car licensed for automatic driving in the streets and highways of the German states, Berlin and Brandenburg. The objective of the project is to develop technology that can be transferred to driver assistance systems, to innovative safety systems for conventional cars, or to full autonomous vehicles in private enclosures such as airports or mines.

The car extracts its position on the Berlin streets from a very accurate GPS unit and a map of the city. Three laser scanners at the front, and